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Report

Rates for breast cancer characteristics by estrogen and progesterone receptor status in the major racial/ethnic groups

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Summary

It has been reported that age-specific breast cancer rates vary by estrogen receptor and progesterone receptor status. We report breast cancer rates for age-at-diagnosis, stage-at-diagnosis, histological grade and type by estrogen (ER) and progesterone (PgR) receptor status in six major racial/ethnic groups. The average annual age-adjusted rates for breast cancers with estrogen receptor positive (ER^+), ER^-, progesterone receptor positive (PgR^+), PgR^-, ER^+PgR^+, ER^+PgR^-, ER^-PgR^+ and ER^-PgR^- are determined from 123,732 breast cancers with known ER status, diagnosed from 1992 to 1998 from 11 Surveillance, Epidemiology, and End Results (SEER) cancer registries. For each racial/ethnic group, their ER^+ (ER^+PgR^+ and ER^+PgR^-) age-specific rates increased with age (but at a slower pace after ages 50–54) while their ER^- (ER^- PgR^+ and ER^-PgR^-) age-specific rates did not increase after ages 50–54. The rank orders of the rates among the racial/ethnic groups varied by ER/PgR status. The stage I rates were greater than the stage II rates for the ER/PgR groups except for ER^- and ER^-PgR^- cancers. The grade 2 (moderately differentiated) rates were greater than the grades 3 and 4 (poorly differentiated and undifferentiated cancers) rates for ER^+ cancers, but not for ER^- cancers. These results suggest that although breast cancer is a disease with enormous heterogeneity, the multiple types of breast cancer can be separated into distinct subgroups by their ER status, and perhaps by their ER/PgR status, and their cancer characteristics may be important in understanding the multiple nature of breast cancer.

Introduction

The influence of estrogen receptor status on the clinical outcomes of breast cancer has been established. That is, the estrogen receptor positive (ER^+) cancers are different from the estrogen receptor negative (ER^-) cancers for predictive factors for treatment [1, 2] and for prognostic factors for survival [1–3]. The clinical evidence is building that the four joint estrogen and progesterone receptors, that is, estrogen receptor positive and progesterone receptor positive (ER^+PgR^+), ER^+PgR^-, ER^-PgR^+ and ER^-PgR^-, are also important. For example, in treating breast cancer recurrence with tamoxifen, there is a gradient in the reduction in recurrence and in mortality by ER/PgR groups with ER^+PgR^+ and ER^+PgR^- having similar reductions followed by ER^-PgR^+ having smaller reductions and ER^-PgR^- having almost no response [4]. In addition, a study of proliferative activity showed that ER/PgR status correlated with the amount of S-phase fraction, a measure of proliferative activity. The S-phase fraction increased significantly from ER^+PgR^+ to ER^+PgR^- to ER^-PgR^+, with ER^-PgR^- having the highest S-phase fraction and the greatest proliferative activity and ER^+PgR^+ cancers having the least [5]. Thus, the importance of ER/PgR status is being established for the treatment and prognosis of breast cancers.

More recently, the role of estrogen receptor status on the formation of breast cancers has been uncovered. It has been demonstrated that ER^- and ER^+ groups as well as the four ER/PgR groups have different age-specific incidence rate patterns in Danish women [6] and in US white and black women [7]. The ER^-

cancers have age-specific rates that level off after age 50 while ER^+ cancer rates increase with age. Thus, breast cancer rates may be composed of at least two different age-specific rate patterns.

The frequency distributions of the major cancer characteristics of the ER/PgR groups have been reported for a number of racial/ethnic groups [8]. It was shown that the frequency distributions of these cancer characteristics vary by the major hormonal groups within each racial/ethnic group. However, these distributions do not allow comparisons between racial/ethnic groups, which require the computation of their age-adjusted rates. In this paper, the average annual age-adjusted breast cancer incidence rates (based on 1992–98 data) by estrogen and progesterone status are reported for age-at-diagnosis, American Joint Committee on Cancer (AJCC) stage, histological grade, and histological type in the major racial/ethnic groups (i.e., white, white non-Hispanic (WNH), black, Hispanic, Asian American and Pacific Islander (AA/PI) and American Indian and Alaska Native (AI/AN) females).

Materials and methods

The data were obtained from population-based data collected by the Surveillance, Epidemiology and End Results Program of the National Cancer Institute (NCI) for the period from 1992 to 1998 [9]. The major racial/ethnic groups have their cancer rates determined annually and include white, white non-Hispanic (WNH), black, Hispanic, Asian American and Pacific Islander (AA/PI) and American Indian, Aleutian Islander and Eskimo, referred to as American Indian and Alaska Native (AI/AN), female breast cancer cases. The Hispanic group is composed of multiple racial groups. As a consequence, the Hispanic group is not mutually exclusive of the other racial groups. The removal of white Hispanics from the white data yields the rates for white non-Hispanics. The invasive breast cancers (i.e., excluding *in situ* cases) used in this study were diagnosed from 1992 through 1998 among residents of 11 geographic areas: Connecticut, Hawaii, Iowa, New Mexico, Utah, Atlanta GA, Detroit MI, Seattle-Puget Sound WA, San Francisco-Oakland CA, San Jose-Monterey CA and Los Angeles CA [9].

The rates are age-adjusted to the 1970 US standard population. This adjustment allows comparison of the rates between the racial/ethnic groups since differences in the populations of the racial/ethnic groups have been standardized. All the rates are averages of the annual 1992–98 rates, unless otherwise specified. The age-specific rates are grouped into three age categories, 0–49, 50–64, and 65 and older (65+). The rates for ages 50–64 are compared to the rates for ages 65+ to determine if rates for the older group are increasing. In addition, the average age-specific rates in 5-year age groups for ages 25–84 are plotted. In the age-specific plots, only white non-Hispanics are presented since the white rates have been previously reported [7].

The ER and PgR status were coded by the SEER Program data collectors based on laboratory results in the medical records at the time of data collection since 1990. The data are reported as positive, negative, unknown (unknown includes borderline/undetermined; ordered but results not in chart; or unknown for ER status) and not done in this report [10].

Data on stage-at-diagnosis are collected according to the SEER Extent of Disease – codes and coding instruction manual [10]. The stage-at-diagnosis used in this report is based on the AJCC's classification of tumors with stages I, II, and III and IV combined [11].

The histological grade codes are determined using the SEER protocol for the collection of these data [10]. Grade 1 is well differentiated, or differentiated, NOS. Grade 2 is moderately differentiated, moderately well differentiated, or intermediate differentiation. Grade 3 is poorly differentiated, and grade 4 is undifferentiated or anaplastic. Grades 3 and 4 were combined in this report.

Histology data have been collected by SEER since 1973. The SEER Program codes both site and histological type according to the second edition of the *International Classification of Diseases for Oncology* (ICD-O-2) [12]. The histological types are grouped according to the recommendations of Simpson and Page [13].

Results

The number and percentages of breast cancers by ER and PgR status and racial/ethnic group, including unknown ER status and ER not done, are reported in Table 1. The average annual adjusted rates for ER^+, ER^-, PgR^+, PgR^-, and the four ER/PgR groups for all the major racial/ethnic groups are reported

Table 1. Numbers and percentages of breast cancers by estrogen and progesterone receptor status from 1992 to 1998

	Total	ER^+	%	ER^-	%	PgR^+	%	PgR^-	%	ER status unknown	%	ER not done	%
White	137,210	81,711	60	23,799	17	68,775	50	33,038	24	24,301	18	7,399	5
White non-Hispanic	126,041	76,326	61	21,399	17	64,135	51	30,144	24	16,522	13	6,753	5
Black	13,634	5,652	41	3,639	27	4,792	35	4,166	31	3,208	24	1,135	8
Hispanic	10,712	5,176	48	2,320	22	4,450	42	2,793	26	2,242	21	620	6
AAPI	10,757	6,414	60	2,202	20	5,681	53	2,732	25	1,609	15	532	5
AI/AN	382	210	55	105	27	183	48	126	33	42	11	25	7

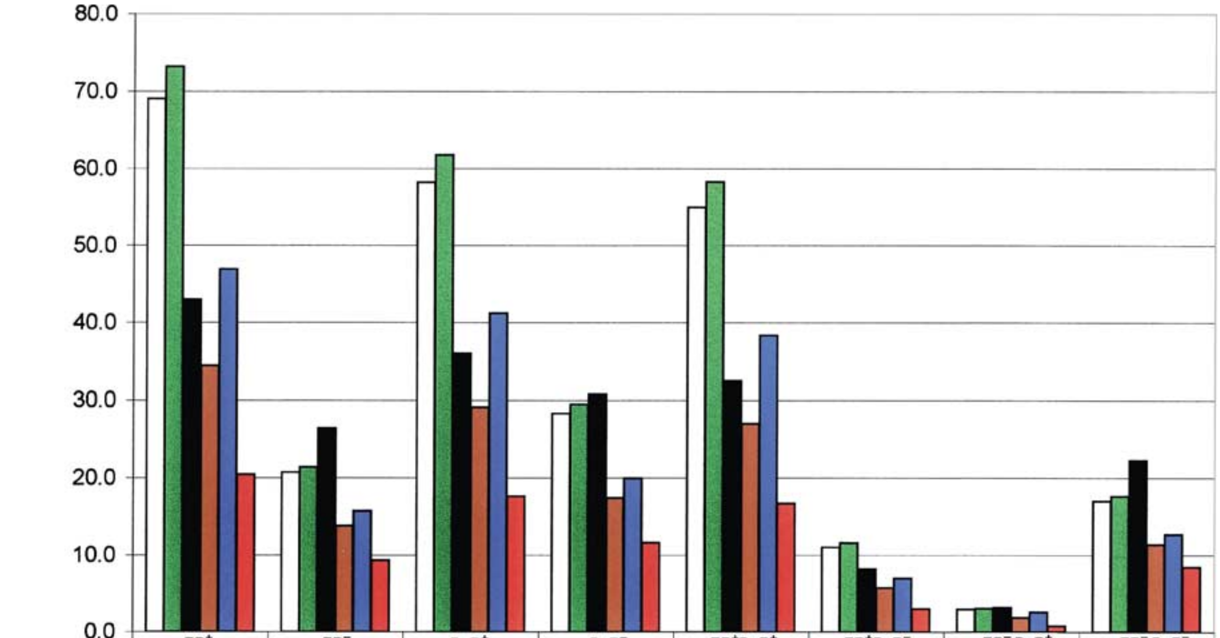
ER/PgR status (Percentages based on total breast cancers)

	Total	ER^+PgR^+	%	ER^+PgR^-	%	ER^-PgR^+	%	ER^-PgR^-	%
White	137,210	64,982	47	13,167	10	3,416	2	19,575	14
White non-Hispanic	126,041	60,719	48	12,285	10	3,071	2	17,592	14
Black	13,634	4,290	31	1,071	8	456	3	3,053	22
Hispanic	10,712	4,090	38	849	8	330	3	1,917	18
AAPI	10,757	5,270	49	943	9	381	4	1,764	16
AI/AN	382	173	45	30	8	nr	nr	95	25

ER/PgR status based on total with known ER/PgR status

	Total	ER^+PgR^+	%	ER^+PgR^-	%	ER^-PgR^+	%	ER^-PgR^-	%
White	101,140	64,982	64	13,167	13	3,416	3	19,575	19
White non-Hispanic	93,667	60,719	65	12,285	13	3,071	3	17,592	19
Black	8,870	4,290	48	1,071	12	456	5	3,053	34
Hispanic	7,186	4,090	57	849	12	330	5	1,917	27
AAPI	8,358	5,270	63	943	11	381	5	1,764	21
AI/AN	307	173	56	30	10	nr	nr	95	31

nr = Not reported, less than 10 cancers.
AA/PI = Asian Americans and Pacific Islanders.
AI/AN = American Indians and Alaska Natives (Aleutian Islander or Eskimo).



	ER^+	ER^-	PgR^+	PgR^-	ER^+PgR^+	ER^+PgR^-	ER^-PgR^+	ER^-PgR^-
White	69.0	20.7	58.3	28.3	55.0	11.0	3.0	17.0
White non-Hisp	73.2	21.4	61.8	29.5	58.3	11.6	3.1	17.6
Black	43.0	26.4	36.0	30.8	32.5	8.2	3.2	22.2
Hispanic	34.4	13.8	29.1	17.4	27.0	5.8	1.9	11.4
AA/PI	46.9	15.7	41.2	19.9	38.4	7.1	2.6	12.7
AI/AN	20.4	9.3	17.6	11.6	16.7	3.1	0.8	8.5

Figure 1. The average annual age-adjusted rates from 1992 to 1998 for estrogen receptor positive (ER^+), estrogen receptor negative (ER^-), progesterone receptor positive (PgR^+), progesterone receptor negative (PgR^-), estrogen receptor positive/progesterone receptor positive (ER^+PgR^+), estrogen receptor positive/progesterone receptor negative (ER^+PgR^-), estrogen receptor negative/progesterone receptor positive (ER^-PgR^+) and estrogen receptor negative/progesterone receptor negative (ER^-PgR^-) groups by the major racial/ethnic groups.

in Figure 1. In Tables 2–5, rates for breast cancer characteristics are reported by ER^+, ER^-, and the four ER/PgR groups for each of the racial/ethnic groups. The age-specific rates for the three age groups (0–49, 50–64, and 65+) as well as the ratios of rates for ages 50–64 to ages 0–49 and ratios of rates of 65+ to 50–64 are reported in Table 2. The 5-year age-specific rates for ages 25–84 for ER^+ and ER^- are reported in Figure 2 and for the four ER/PgR groups in Figure 3 for white non-Hispanics, blacks, Hispanics, AA/PI and AI/AN females. The average age-adjusted rates for AJCC stages-at-diagnosis I, II, and III and IV and the ratios of rates for stage I to stage II are reported in Table 3. The age-adjusted rates for the histological grades, 1, 2, and 3 and 4 and the ratios of rates of grade 2 to grade 3 and 4 are reported in Table 4. The histological types are reported in Table 5.

Rank order of rates of the racial/ethnic groups by estrogen and progesterone status

From Figure 1, for each racial/ethnic group, ER^+PgR^+ rates are greater than (>) ER^-PgR^- rates > ER^+PgR^- rates > ER^-PgR^+ rates. For total breast cancer incidence, white non-Hispanic breast cancer rates > white > black > AA/PI > Hispanic > AI/AN rates (data not shown). But the rank order changes for the different ER/PgR groups. For ER^+ and ER^+PgR^+ breast cancers, white non-Hispanic breast cancer rates > white > AA/PI > black > Hispanic > AI/AN rates. That is, for ER^+ cancers, AA/PI have the second highest rates relative to whites followed by black females. For ER^- and ER^-PgR^- groups, black breast cancer rates > white non-Hispanics > white > AA/PI > Hispanic > AI/AN rates. That is, for ER^- cancers, blacks have the highest rates followed by whites. For

Table 2. Average annual age-specific breast cancer incidence rates by estrogen and progesterone receptor status for the major racial/ethnic groups from 1992 to 1998

Race	Age at diagnosis	ER^+	ER^-	ER^+PgR^+	ER^+PgR^-	ER^-PgR^+	ER^-PgR^-
White	0–49	16.5	8.4	13.8	2.0	1.4	6.7
	50–64	190.9	59.3	151.1	30.8	8.3	49.0
	65+	290.8	57.7	227.7	50.8	7.5	48.2
	Ratio 50–64/0–49	11.6	7.1	10.9	15.4	5.9	7.3
	Ratio 65+/50–64	1.5	1.0	1.5	1.6	0.9	1.0
White non-Hispanic	0–49	17.7	8.6	14.8	2.1	1.4	6.9
	50–64	204.4	61.8	162.1	32.7	8.7	51.0
	65+	303.7	59.5	237.9	53.0	7.7	49.7
	Ratio 50–64/0–49	11.5	7.2	11.0	15.6	6.2	7.4
	Ratio 65+/50–64	1.5	1.0	1.5	1.6	0.9	1.0
Black	0–49	11.3	11.3	8.9	1.9	1.6	9.3
	50–64	115.0	77.6	85.6	23.0	9.5	65.1
	65+	178.8	66.2	134.3	35.0	6.2	57.5
	Ratio 50–64/0–49	10.2	6.9	9.6	12.1	5.9	7.0
	Ratio 65+/50–64	1.6	0.9	1.6	1.5	0.7	0.9
Hispanic	0–49	9.2	6.2	7.6	1.2	1.0	5.1
	50–64	91.6	38.3	70.8	16.1	5.4	31.6
	65+	142.4	35.1	110.3	26.0	4.1	30.1
	Ratio 50–64/0–49	10.0	6.2	9.3	13.5	5.4	6.2
	Ratio 65+/50–64	1.6	0.9	1.6	1.6	0.8	1.0
AA/PI	0–49	15.6	6.6	13.4	1.6	1.5	4.9
	50–64	132.2	47.7	105.9	21.3	7.1	39.3
	65+	160.0	38.1	128.8	27.4	4.8	32.8
	Ratio 50–64/0–49	8.5	7.2	7.9	13.3	4.7	8.0
	Ratio 65+/50–64	1.2	0.8	1.2	1.2	0.7	0.8
AI/AN	0-49	5.9	3.6	5.2	nr	nr	3.2
	50–64	56.4	37.6	44.5	10.0	nr	34.7
	65+	78.2	11.2	63.4	nr	nr	9.9
	Ratio 50–64/0–49	9.6	10.4	8.6			10.8
	Ratio 65+/50–64	1.4	0.3	1.4			0.3

nr = not reported less than 10 cancers in group.
AA/PI = Asian Americans and Pacific Islanders.
AI/AN = American Indians and Alaska Natives (Aleutian Islander or Eskimo).

ER^+PgR^- cancers, the order is the same as for the total breast cancer rates. The rates for ER^-PgR^+ are low, and it is difficult to make comparisons. Thus, the rank orders of the rates of the racial/ethnic groups vary by the ER/PgR groups.

Age-specific rates by estrogen and progesterone receptor status

Inspection of Figures 2 and 3 indicates that ER^+ (ER^+PgR^+ and ER^+PgR^-) rates continue to increase after age 50–54 but at a slower pace. In contrast, the age-specific rates are not increasing for ER^- (ER^-PgR^+ and ER^-PgR^-) groups after ages 50–54. These observations are confirmed in examining rates in Table 2. The changes in the rates prior to age 50, as measured by the ratios of the 50–64 rate to the 0–49 rate, are greater than the changes in rates after age 50, as measured by the ratios of the 65+ rate to the 50–64 rate. Furthermore, ER^+, ER^+PgR^+, and ER^+PgR^- groups have age-specific incidence rates that increase with older age (i.e., 65+ rates >50–64 rates in Table 2)

Table 3. Average annual age-adjusted breast cancer incidence rates by AJCC stage-at-diagnosis and by estrogen and progesterone receptor status for the major racial/ethnic groups from 1992 to 1998

Race	AJCC stage	ER^+	ER^-	ER^+PgR^+	ER^+PgR^-	ER^-PgR^+	ER^-PgR^-
White	I	35.9	7.9	28.9	5.3	1.3	6.3
	II	24.5	9.0	19.5	4.0	1.2	7.5
	III&IV	5.8	2.7	4.4	1.2	0.4	2.3
	Ratio I/II	1.5	0.9	1.5	1.3	1.1	0.8
White non-Hispanic	I	38.4	8.3	30.9	5.7	1.3	6.7
	II	25.7	9.2	20.5	4.2	1.2	7.7
	III&IV	6.1	2.8	4.6	1.2	0.4	2.4
	Ratio I/II	1.5	0.9	1.5	1.4	1.1	0.9
Black	I	17.2	8.0	13.0	3.3	1.0	6.8
	II	17.8	12.2	13.8	3.1	1.5	10.3
	III&IV	5.5	4.7	3.8	1.4	0.5	4.0
	Ratio I/II	1.0	0.7	0.9	1.1	0.7	0.7
Hispanics	I	15.3	4.3	12.1	2.5	0.8	3.4
	II	14.2	6.7	11.1	2.5	0.8	5.7
	III&IV	3.5	2.0	2.7	0.7	0.2	1.7
	Ratio I/II	1.1	0.6	1.1	1.0	1.0	0.6
AA/PI	I	23.8	6.0	19.6	3.5	1.1	4.8
	II	18.0	7.0	14.7	2.6	1.1	5.7
	III&IV	3.5	2.1	2.7	0.7	0.3	1.7
	Ratio I/II	1.3	0.9	1.3	1.3	1.0	0.8
AI/AN	I	9.1	3.6	8.0	1.0	0.5	3.1
	II	7.6	3.4	6.1	1.2	0.2	3.2
	III&IV	2.7	1.8	1.6	0.8	0.2	1.6
	Ratio I/II	1.2	1.1	1.3	0.8	2.8	1.0

AA/PI = Asian Americans and Pacific Islanders.
AI/AN = American Indians and Alaska Natives (Aleutian Islander and Eskimo).

except for ER^+PgR^- cancers in American Indians and Alaska Natives. In contrast, ER^-, ER^-PgR^-, and ER^-PgR^+ groups have age-specific rates that do not increase after ages 50–64 (50–64 rates > 65+ rates in Table 2). Thus, all the racial/ethnic groups have the same patterns for their age-specific incidence rates that increase with older age for ER^+ cancers but at a slower pace after age 50 and do not increase with older age for ER^- cancers.

Further examination of the 5-year age-specific rates by racial/ethnic group in Figures 2 and 3 reveals some interesting observations, not available from the overall age-adjusted rates. We highlight two examples. First, ER status divides breast cancer rates so that ER^+ rates increase with older age and ER^- rates do not. In contrast, PgR status does not divide its groups into these unique patterns, indicating that ER status is better able to identify unique breast cancer subtypes than PgR status. Secondly, for ER^- cancers, black rates are the highest from ages 25 to 84. Thus, for ER^- cancers, there is no crossover point where white rates are greater than black rates for this age range.

Rates by stage-at-diagnosis and estrogen and progesterone receptor status

For AJCC stage data (Table 3), stages I and II have the highest incidence rates. The stage I > stage II rates for ER^+ and ER^+PgR^+ groups (ratios of stage I/II > 1) for the racial/ethnic groups except for black females. In contrast, the rates for stage II ≥ stage I for ER^- and ER^- PgR^- groups (ratio of stage I/II < 1). For all the racial/ethnic groups, the ratios of stage I/II are greatest for ER^+ and ER^+PgR^+, followed by ER^+PgR^-, then

Table 4. Average annual age-adjusted breast cancer incidence rates by histological grade and by estrogen and progesterone receptor status for the major racial/ethnic groups from 1992 to 1998

Race	Grade	ER^+	ER^-	ER^+PgR^+	ER^+PgR^-	ER^-PgR^+	ER^-PgR^-
White	1	11.8	0.9	9.6	1.6	0.3	0.6
	2	27.7	4.3	22.6	3.9	0.8	3.3
	3&4	17.3	12.4	13.2	3.4	1.4	10.6
	Ratio 2/3&4	1.6	0.3	1.7	1.1	0.6	0.3
White non-Hispanic	1	12.6	0.9	10.3	1.7	0.3	0.6
	2	29.4	4.5	24.0	4.1	0.9	3.5
	3&4	18.2	12.8	13.9	3.6	1.4	10.9
	Ratio 2/3&4	1.6	0.4	1.7	1.1	0.6	0.3
Black	1	5.8	0.7	4.5	1.0	0.2	0.5
	2	15.2	4.0	11.5	2.8	0.7	3.2
	3&4	12.5	17.2	9.1	2.8	1.8	14.8
	Ratio 2/3&4	1.2	0.2	1.3	1.0	0.4	0.2
Hispanic	1	5.5	0.5	4.4	0.7	0.2	0.3
	2	13.3	2.4	10.8	2.0	0.4	1.9
	3&4	9.5	9.0	7.2	2.0	1.0	7.7
	Ratio 2/3&4	1.4	0.3	1.5	1.0	0.4	0.2
AA/PI	1	7.2	0.7	6.1	0.8	0.2	0.4
	2	19.6	3.7	16.3	2.7	0.8	2.8
	3&4	12.6	9.2	10.1	2.3	1.2	7.8
	Ratio 2/3&4	1.6	0.4	1.6	1.2	0.7	0.4
AI/AN	1	2.5	0.2	2.3	0.2	0.1	0.1
	2	8.8	1.4	7.7	0.9	0.2	1.2
	3&4	4.8	6.3	3.6	1.2	0.4	5.9
	Ratio 2/3&4	1.8	0.2	2.2	0.7	0.5	0.2

AA/PI = Asian Americans and Pacific Islanders.
AI/AN = American Indians and Alaska Natives (Aleutian Islander and Eskimo).
Grade 1: Well differentiated.
Grade 2: Moderately differentiated.
Grade 3: Poorly differentiated.
Grade 4: Undifferentiated.

ER^-PgR^+ and finally ER^- and ER^-PgR^- cancers. Only ER^-PgR^- cancers have consistently more stage II than stage I cancers for each racial/ethnic group. Thus, the stage I rates were greater than the stage II rates for the ER/PgR groups except ER^- and ER^-PgR^- cancers.

Rates by histological grade and estrogen and progesterone receptor status

For histological grade (Table 4), the highest rates are for grade 2 (moderately differentiated) and grade 3 (poorly differentiated) with grade 4 (undifferentiated) cancers. For ER^+, ER^+PgR^+ and ER^+PgR^- groups, grade 2 > grades 3 and 4 rates (ratio grade 2/3 and 4 > 1) except for ER^+PgR^- for AI/ANs. For ER^-, ER^-PgR^+ and ER^-PgR^- groups, grade 3 and 4 > grade 2 rates (ratio grade 2/3 and 4 < 1). Thus in general, the grade 2 rates were greater than the grades 3 and 4 rates for ER^+ cancers, but not for ER^- cancers.

Rates by histological type and estrogen and progesterone receptor status

For histological type, the predominant cell type is infiltrating duct carcinoma for each ER/PgR group of every racial/ethnic group (Table 5). However, the

Table 5. Average annual age-adjusted breast cancer incidence rates by histological type and by estrogen and progesterone receptor status for the major racial/ethnic groups from 1992 to 1998

Race	Histology	ER^+	ER^-	ER^+PgR^+	ER^+PgR^-	ER^-PgR^+	ER^-PgR^-
White	Duct	51.5	16.8	41.3	8.1	2.3	13.9
	Lobular	7.3	0.8	5.6	1.3	0.2	0.6
	Tubular	1.5	0.2	1.2	0.2	0	0.1
	Mucinous	1.9	0.1	1.5	0.3	0	0.1
	Medullary	0.2	0.8	0.1	0.1	0.1	0.7
	Comedoca	1.0	0.8	0.7	0.2	0.1	0.6
White Non-Hispanic	Duct	54.5	17.4	43.8	8.5	2.4	14.5
	Lobular	7.8	0.9	6.1	1.4	0.2	0.6
	Tubular	1.6	0.2	1.3	0.3	0	0.1
	Mucinous	2.0	0.2	1.6	0.3	0	0.1
	Medullary	0.2	0.7	0.1	0.1	0	0.7
	Comedoca	1.1	0.8	0.8	0.2	0.1	0.7
Black	Duct	32.6	22.2	24.6	6.3	2.6	18.8
	Lobular	3.5	0.6	2.6	0.6	0.1	0.4
	Tubular	0.5	0.1	0.3	0.1	0	0.1
	Mucinous	1.5	0.1	1.1	0.3	0	0.1
	Medullary	0.3	1.4	0.2	0.1	0.1	1.2
	Comedoca	1.1	0.9	0.8	0.2	0.2	0.7
Hispanic	Duct	26.2	11.0	20.6	4.4	1.6	9.1
	Lobular	3.1	0.5	2.3	0.7	0.1	0.4
	Tubular	0.5	0.1	0.4	0.1	0	0
	Mucinous	1.1	0	0.9	0.2	0	0
	Medullary	0.2	0.8	0.1	0.1	0.1	0.7
	Comedoca	0.6	0.6	0.4	0.1	0.1	0.5
AA/PI	Duct	38.0	13.0	31.3	5.6	2.2	10.5
	Lobular	2.5	0.3	2.1	0.4	0.1	0.3
	Tubular	0.6	0.1	0.5	0.1	0	0.1
	Mucinous	1.9	0.1	1.5	0.3	0	0.1
	Medullary	0.2	0.6	0.1	0.1	0	0.5
	Comedoca	1.1	0.8	0.9	0.2	0.1	0.6
AI/AN	Duct	16.8	7.4	13.7	2.6	0.6	6.8
	Lobular	0.9	0.2	0.7	0	0	0.2
	Tubular	0.2	0	0.1	0.1	0	0
	Mucinous	0.5	0	0.5	0	0	0
	Medullary	0.1	0.7	0	0.1	0	0.7
	Comedoca	0.5	0.5	0.4	0	0.1	0.4

Duct = Infiltrating duct carcinoma = Infiltrating duct carcinoma includes infiltrating duct carcinoma (8500), adenocarcinoma, NOS (8140), carcinoma, NOS (8010), epithelimoa, NOS (8011), scirrhus adenocarcinoma (8141), papillary carcinoma, NOS (8050), papillary adenocarcinoma, NOS (8260), intraductal papillary adenocarcinoma with invasion (8503), inflammatory carcinoma (8530).
Lobular = Infiltrating lobular carcinoma = Infiltrating lobular carcinoma, NOS (8520), Infiltrating ductular carcinoma (8521).
Tubular = Tubular carcinoma = Adenoid cystic carcinoma (8200), cribriform carcinoma (8201), tubular adenocarcinoma (8211).
Mucinous = Mucinous carcinoma (colloid) carcinoma = Mucinous adenocarcinoma/colloid carcinoma (8480), Mucin-producing adenocarcinoma (8481).
Medullary = Medullary carcinoma = Medullary, carcinoma NOS (8510), Medullary carcinoma with amyloid stroma (8511), Medullary carcinoma with lymphoid stroma (8512).
Comedoca = Comedocarcinoma = Comedocarcinoma, NOS (8501).

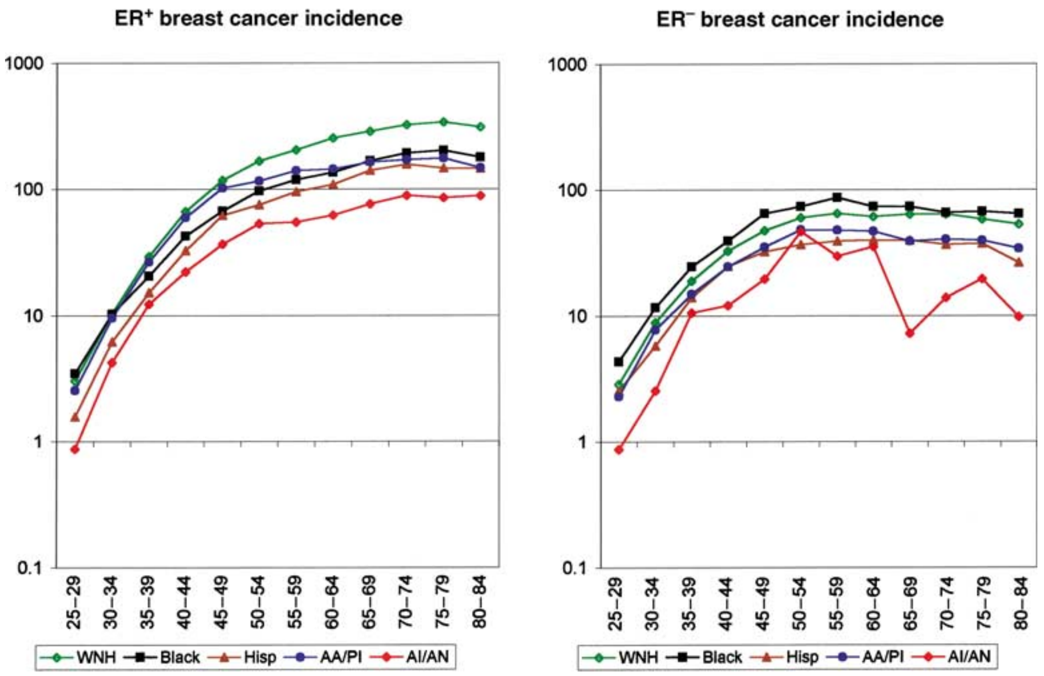


Figure 2. The average 5-year age group specific rates for ages 25–84 for ER^+ and ER^- cancers by major racial/ethnic group, based on breast cancers diagnosed from 1992 to 1998.

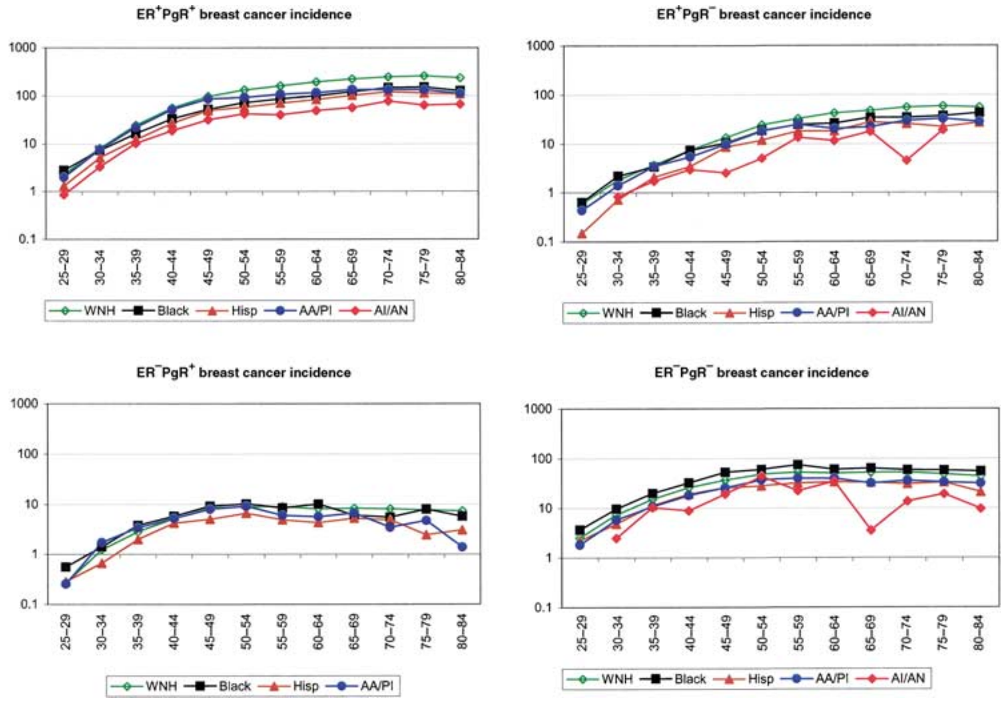


Figure 3. The average 5-year age group specific rates for ages 25–84 for ER^+PgR^+, ER^+PgR^-, ER^-PgR^+ and ER^-PgR^- cancers by major racial/ethnic group, based on breast cancers diagnosed from 1992 to 1998.

second and third leading histological types vary by ER group. For ER^+ tumors (ER^+PgR^+, ER^+PgR^-), infiltrating lobular carcinoma and mucinous carcinoma are a distant second and third most frequent, respectively, for the racial/ethnic groups. For ER^- and ER^-PgR^- groups, infiltrating lobular carcinoma, medullary carcinoma and comedocarcinoma are the next most frequent histological types for the racial/ethnic groups. For ER^-PgR^+ cancers, there are too few cancers to assess additional histological types for any racial/ethnic group. Thus, the overwhelming histology is infiltrating duct carcinoma for each ER/PgR group.

Discussion

The findings that the age-specific breast cancer rates are composed of at least two distinctly different age-specific rate patterns, one for ER^+ cancers and another for ER^- cancers that may be further stratified by the four ER/PgR groups, have been confirmed in our examination of the major racial/ethnic groups in SEER. The ER^+(ER^+PgR^+ and ER^+PgR^-) cancer rates increase with age but at a slower pace after age 50–54 while the ER^- (ER^-PgR^+ and ER^-PgR^-) cancer rates do not increase after age 50–54 in each racial/ethnic group examined. Thus, the observations that the age-specific breast cancer rates are composed of a number of different age-specific distributions, defined by ER status and PgR status, appears to be an universal phenomenon among the major racial/ethnic groups in the US. These results suggest that breast cancer may need to be examined separately by ER^+ and ER^- cancers in future etiological and clinical studies.

Possible mechanisms for ER^- and ER^+ formation

Since the changes in the two age-specific rate distributions for ER status occur around the age of 50, the role of menopause in the changes in slopes is strongly suggested [6, 7]. However, there is an apparent paradox [6]. Our results and those of others [6, 7] indicate that the ER^- cancers (which would be expected to be less sensitive to hormonal changes) appear to be very sensitive to the changes in estrogen levels at menopause. The risk is level after menopause while the ER^+ cancers (which are thought to be sensitive to estrogen levels) are less sensitive to the hormonal changes at menopause since the risk still increases with older age, albeit at a slower pace. Tarone and Chu have proposed a possible explanation for these results [7]. In normal breast tissue, normal cell proliferation is caused by the paracrine mechanism of normal ER^+ cells combining with estrogen to trigger cell proliferation of normal ER^- cells [14–17]. That is, normal cell proliferation occurs in normal ER^- cells that are triggered to divide by normal ER^+ cells combining with estrogen, and does not involve normal ER^+ cells dividing [14–17]. If the formation of pre-cancerous ER^- lesions follows this pathway, then the loss of high levels of estrogen at menopause will stop the formation of the pre-cancerous ER^- lesions, freezing the risk at the initial menopausal levels which generally occur by ages 50–54.

Normal ER^+ cells have been shown to increase with age after menopause. These increased ER^+ cells can react with lower levels of estrogen to cause ER^- cell proliferation, but at reduced rates since the estrogen levels are lower. Recently, Fuqua and colleagues have shown that an estrogen receptor-α mutation in pre-malignant breast lesions causes a hypersensitivity to estrogen [18]. These hypersensitive ER^+ cells can react with lower levels of estrogen after menopause to cause increases in ER^+ cancers. However, the hypersensitivity may not completely overcome the sharp fall in estrogen with menopause, creating the slower increase in ER^+ rates after menopause. Thus, if the paracrine mechanism accounts for the formation of both ER^- and ER^+ cancers then the ER/PgR groups may be due to a single mechanism of action with a spectrum of clinical manifestations. Accordingly, the risk factors for ER^+ and ER^- and the four ER/PgR groups, such as effects of age-at-first child-birth and nulliparity, would be expected to be the same if the paracrine mechanism is operating for both ER^- and ER^+ formation.

However, other possibilities for ER^+ formation may exist, such as activation of ER^+ cell proliferation after menopause. If there is more than a single mechanism for ER^+ and ER^- breast cancer formation, such as a mechanism for ER^- and other mechanisms for ER^+, then the risk factors for ER^+ and ER^- and even for the four ER/PgR groups may be different.

Initial epidemiological and molecular studies suggest that ER^+ (ER^+PgR^+ and ER^+PgR^-) groups may be different from ER^- (ER^-PgR^+ and ER^-PgR^+) groups. In epidemiological studies, risk factor patterns were different for ER^+PgR^+ and ER^-PgR^- cancers. For example, older age at menarche was a protective factor for ER^+PgR^+ but had no effect on ER^-PgR^-

cancers [19–21]. Nulliparity appears to be a risk factor for ER^+PgR^+ cancers but not for ER^-PgR^- cancers [19, 21]. Future epidemiological studies of the risk factors by the four ER/PgR groups will confirm if the ER/PgR groups have different risk factors. These findings will provide important insights into the potential mechanisms of breast cancer formation.

On the molecular side, a microarray study of over 1,750 genes in 84 experimental samples indicated that the gene characteristics can be divided into two large groups, one for ER^+ and another for ER^- [22]. The ER+ cancers were characterized by relatively high expression of many genes expressed by breast luminal cells, with little to no *Erb-B2* expression at high levels. At this molecular level, the ER^- cancers are characterized by two subgroups one involving basal epithelial cells and another involving high *Erb-B2* expression, further subdividing ER^- cancers. If these differences lead to different mechanisms of breast cancer formation, then the risk factors by ER status and perhaps PgR status would be expected to be different.

One important example of the impact of ER status on breast cancer is in the comparison of black and white breast cancer incidence rates. For overall breast cancer incidence, black rates are higher than white rates until age 40–44, when white rates become higher. However, examination of the rates by ER status indicates that for ER^- cancers, black rates have the highest rates for ages 25–84. In contrast for ER^+ cancers, the black rates are higher until ages 35–39. Thus the black/white crossover at age 40–44 for all breast cancers was simply an artifact of the combining of the age-specific rates for the two types of ER cancers. This example illustrates the importance of considering ER^+ and ER^- cancers separately in studies involving breast cancer formation.

Possible explanations for black incidence rates being greater than white rates

For both ER^+ and ER^- cancers in younger women (ages <35) and for ER^- cancers for ages 25–84, black incidence rates are greater than white rates. There are three possible sources for the higher rates in black females. One possible explanation would be greater medical care and consequently more early detection for black females. Given the higher mortality rates for young black females [23], it is unlikely that this is explanation for the higher rates. Another possible explanation is that black females have a greater exposure to risk factors than white females early in their lives, including possible *in utero* exposures. Examination of known risk factors in young black women with breast cancer in a recent case-control study found little association of established risk factors with disease [24]. However, the study did not analyze the risk factors by ER status. Furthermore, the role of the second estrogen receptor site, ER-β, is still under active investigation [25]. Finally, the higher rates in black females may have a genetic link. The observations that some hereditary breast cancers, such as BRCA1 cancers, have been associated with more ER^- cancers [26] and black females have more ER^- cancers may be more than coincidental. On the other hand, other hereditary breast cancers, such as BRCA2, are not associated with more ER^- cancers [26]. Clearly, more research is needed to better understand the causes of the higher rates in black females.

Distinction between cancer formation and cancer progression phenomena

The distinctions between cancer formation and cancer progression phenomena are important since our age-specific results (pre-cancerous ER^+ lesions do not covert to pre-cancerous ER^- lesions) may appear contradictory to clinical experience (that hormone responsive (associated with ER^+) cancers can become hormone non-responsive (associated with ER^-) cancers). However, they describe different aspects of the carcinogenic process. In cancer formation, reflected in the age-specific cancer incidence rate patterns, pre-neoplastic ER^+ lesions do not transform to ER^- pre-neoplastic lesions after menopause since ER^- rates do not increase after menopause. That is, the initial formation of pre-neoplastic ER^- lesions is fixed by age 50 and is independent of formation of pre-neoplastic ER^+ lesions after this age. In cancer progression, cancers can progress from hormone dependent cancers to hormonal independent cancers as cancers move from being responsive to tamoxifen to becoming non-responsive to tamoxifen. Some may imply that these changes indicate that ER^+ cancers progress to ER^- cancers, (although others argue that this shift reflects development of drug resistance rather than a change in the biological nature of the patient's cancer) [27]. Thus, the initial formation of ER^- cancers is independent of the initial formation of ER+ cancers, at least after menopause, but once formed, hormone-dependent cancers may progress to hormone-independent cancers.

Limitations to the study

There are some limitations associated with our study using the SEER public-use database [9]. As discussed previously, the ER and PgR data are from 11 cancer registries and performed by many different laboratories. Fortunately, the distributions of the four ER/PgR cancers for white females are similar to those reported in the literature [19, 28]. For the minority groups, ER/PgR data from SEER have not been reported previously in the detail presented in this paper. Second, ER and PgR data for the period 1992–98 have 18% unknowns and 5% not done. If the ER status for these types is not representative of the known ER distributions then some observations about the known ER groups may change. Third, the number of cancers in subgroups may be small for some minorities, leading to some uncertainty about the shape of the age distributions, particularly for AI/ANs. On the other hand, if ER/PgR status is important in breast cancer, there is a need to report as much data as possible on minorities. Finally, data on the second estrogen receptor, ER-β, is not collected by SEER.

Conclusions

The breast cancer incidence rates are composed of at least two different age-specific rate patterns (one for ER^+ and another for ER^-) and as many as four, representing different ER/PgR groups. These results are seen in all the six major racial/ethnic groups examined and appear to be a general result in the SEER Program. Thus, ER status, and perhaps PgR status, impacts not only on treatment and prognosis but also on the formation of breast cancers, their etiology. These results may fundamentally affect how breast cancer studies are analyzed. There may be a need to examine breast cancers by ER^+ and ER^- status separately in etiological as well as clinical studies.

In conclusion, the discovery that ER status and ER/PgR status are important in cancer formation as well as in the prognosis and treatment of breast cancer may indicate that ER and ER/PgR status are fundamentally important in subdividing breast cancers. The cancer characteristics of these groups may provide important keys to understanding the multiple nature of breast cancers.

References

1. Early Breast Cancer Trialists' Collaborative Group: Polychemotherapy for early breast cancer: an overview of the randomised trials. Early Breast Cancer Trialists' Collaborative Group [see comments]. Lancet 352: 930–942, 1998
2. Elledge RM, Fuqua SAW: Estrogen and progesterone receptors. In: Harris JR, Lippman ME, Morrow M, Osborne CK (eds) Diseases of the Breast. 2nd edn, Lippincott Williams & Wilkins, Philadelphia, 2000, pp 471–488
3. Harris JR, Lippman ME, Morrow M, Osborne CK (eds): Diseases of the Breast. 2nd edn, Lippincott Williams & Wilkins, Philadelphia, 2000
4. Early Breast Cancer Trialists' Collaborative Group: Tamoxifen for early breast cancer: an overview of the randomised trials. Early Breast Cancer Trialists' Collaborative Group [see comments]. Lancet 351: 1451–1467, 1998
5. Wenger CR, Beardslee S, Owens MA, Pounds G, Oldaker T, Vendely P, Pandian MR, Harrington D, Clark GM: DNA ploidy, S-phase, and steroid receptors in more than 127,000 breast cancer patients. Breast Cancer Res Treat 28: 9–20, 1993
6. Yasui Y, Potter JD: The shape of age-incidence curves of female breast cancer by hormone-receptor status. Cancer Causes Cont 10: 431–437, 1999
7. Tarone RE, Chu KC: The greater impact of menopause on ER^- than ER^+ cancer incidence: a possible explanation. Cancer Causes Control 13: 7–14, 2002
8. Chu KC, Anderson WF, Fritz A, Ries LG, Brawley: Frequency of breast cancer characteristics by their joint estrogen and progesterone receptor status for multiple racial/ethnic groups. Cancer 92: 37–45, 2000
9. Surveillance, Epidemiology and End Results SEER Program: SEER Cancer Incidence Public Use-Database, 1973–1998. SEER*Stat:(4.0) 2001
10. Shambaugh EM, Ries LAG, Kruse MA, Platz CE, Ryan RF, Weiss MA: SEER Extent of Disease – 1988 Codes and Coding Instructions. 2nd edn, 1992
11. Seiffert JE: SEER Program: Comparative Staging Guide for Cancer. Version 1.1, 1993
12. Percy CL, Holton VV, Muir C: International Classification of Diseases for Oncology. 2nd edn, World Health Organization, Geneva, 1990
13. Simpson JF, Page DL: Status of breast cancer prognostication based on histopathologic data. Am J Clin Pathol 102(suppl): S3–S8, 1994
14. Clarke RB, Howell A, Potten CS, Anderson E: Dissociation between steroid receptor expression and cell proliferation in the human breast. Cancer Res 57: 4987–4991, 1997
15. Anderson E, Clarke RB, Howell A: Estrogen responsiveness and control of normal human breast proliferation. J Mammary Gland Biol Neoplasia 3: 23–35, 1998
16. Shoker BS, Jarvis C, Sibson DR, Walker C, Sloane JP: Oestrogen receptor expression in the normal and pre-cancerous breast. J Pathol 188: 237–244, 1999
17. Russo J, Russo IH: Cellular basis of breast cancer susceptibility. Oncol Res 11: 169–178, 1999
18. Fuqua SAW, Wiltschke C, Zhang QX, Borg A, Castles CG, Friedrichs WE, Hopp T, Hilsenbeck S, Mohsin S, O'Connell P, Allred DC: A hypersensitive estrogen receptor-α mutation in premalignant breast lesions. Cancer Res 60: 4026–4029, 2000
19. Potter JD, Cerhan JR, Sellars TA, McGovern PG, Drinkard C, Kushi LR, Folsom AR: Progesterone and estrogen receptors and mammary neoplasia in the Iowa Women's Health Study: how many kinds of breast cancer are there? Cancer Epidemiol Biomark Prev 4: 319–326, 1995
20. Yoo KY, Tajima K, Miura S, Takeuchi T, Hirose K, Risch H, Dubrow R: Breast cancer risk factors according to

combined estrogen and progesterone receptor status: a case-control analysis. Am J Epidemiol 146: 307–314, 1997
21. Huang WY, Newman B, Millikan RC, Schell MJ, Hulka BS, Moorman PG: Hormone-related factors and risk of breast cancer in relation to estrogen receptor and progesterone receptor status. Am J Epidemiol 151: 703–714, 2000
22. Perou CM, Sorlie T, Eisen MB, van de Rijn M, Jeffrey SS, Rees CA, Pollack JR, Ross DT, Johnsen H, Akslen LA, Fluge O, Pergamenshcikov A, Williams C, Zhu SX, Lonning PE, Borresen-Dale A, Brown P, Botstein D: Molecular portraits of human breast tumors. Nature 406: 747–752, 2000
23. Chu KC, Tarone RE, Brawley: Breast cancer trends for black women compared with white women. Arch Fam Med 8: 521–528, 1999
24. Brinton LA, Benichou J, Gammon MD, Brogan DR, Coates R, Schoenberg JB: Ethnicity and variation in breast cancer incidence. Int J Cancer 73: 349–355, 1997
25. Katzenellenbogen BS, Katzenellenbogen JA: Estrogen receptor transcription and transactivation: estrogen receptor alpha and estrogen receptor beta: regulation by selective estrogen receptor molecules and importance in breast cancer. Breast Cancer Res 2: 335–344, 2000
26. King MC, Wieand S, Hale K, Lee M, Walsh T, Owens K, Tait J, Ford L, Dunn BK, Costantino J, Wickerham L, Wolmark N, Fisher B: Tamoxifen and breast cancer incidence among women with inherited mutations in BRCA1 and BRCA2. JAMA 286: 2251–2256, 2001
27. Robertson JFR: Oestrogen receptor: a stable phenotype in breast cancer. Br J Cancer 73: 5–12, 1996
28. Wittliff JL: Steroid-hormone receptors in breast cancer. Cancer 53: 630–643, 1984

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